

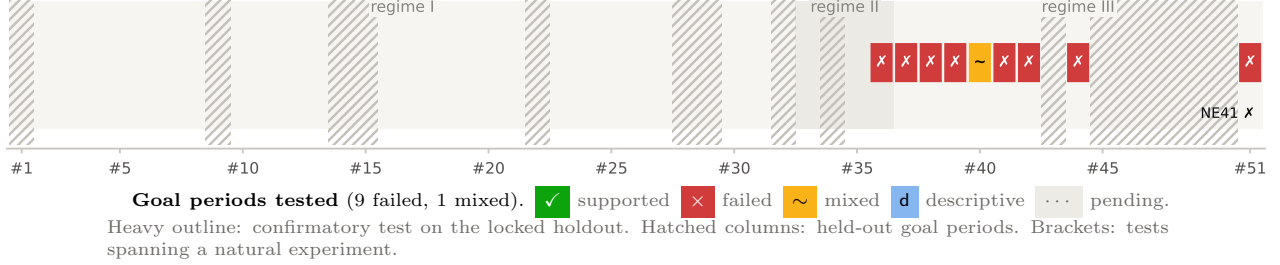
H134: A Pólya-urn self-field holds an agent on its project: the leave odds fall as the project's share of the agent's own context rises

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 45% Faithfulness 0.5/5 Usefulness 1/5

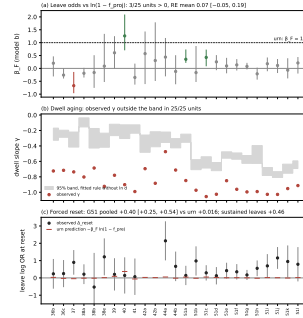
Bottom line. Project stickiness ages strongly, but the context's project self-share does not carry it; a context reset releases the agent as a step, not as the urn's dose.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

The claim (HH377): an agent stays on a project because its own context is full of that project. The field toward the current project is $-\ln(1-f)$, with f the share of the agent's own calls in the context segment that touched the project (a Pólya urn, as H16 and H72 found for idle traps). The urn predicts a leave slope of 1 on $\ln(1-f)$, and its per-call rule should reproduce the observed aging of project dwell without fitting it. On 25 regime-III units (989,450 calls, 58,043 project visits) the slope is $+0.07 [-0.05, +0.19]$, and the rule misses the dwell aging in 25/25 units. The context still matters as a step: a forced reset raises leaving by $+0.40$ log-odds in G51, 24 times the urn's prediction. The reserved data are not used.



Observables. (a) Urn slope β_F per unit with agent-day block-bootstrap 95% CIs; the dashed line is the urn's value 1. (b) Observed dwell slope γ (red) against the 95% band of the fitted per-call rule without dwell (grey, posterior-predictive on the observed risk calls). (c) Leave log odds ratio at the first call after a forced reset, matched on agent, dwell and self-share decile, with the urn's predicted step (red).

PHYSICAL MODEL

Model 10 (kinetic Potts walker) with a self-field, model 02 (escape kinetics), model 08 (self-copying). A visit is a run of calls with one carried project label (shared per-call labels, expiry after 100 own calls). At each own call the agent leaves (touches a new project) or stays. Base terms: agent effects, time in the day, previous call kind, reset at the call, named reads about other projects (K). Rivals: the artifact store holds the project (no reset step), intrinsic aging, per-visit frailty.

$$\text{logit } P(\text{leave}_c) = \alpha_i + z_c \cdot \delta + \beta_F \ln(1 - f_c) + \beta_d \ln d_c + \gamma_R K_c$$

$$f_c = \frac{n_a}{n_{\text{own}} + \frac{1}{2}} \quad (0 \text{ after a reset})$$

$$\text{urn: } \beta_F = 1, \beta_d = 0; \quad \Delta_{\text{reset}} = -\beta_F \ln(1 - f_{\text{pre}})$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A context reset frees an agent from its project, but not in proportion to how much of the context the project fills. In #51 a forced reset raises the leave odds by a factor of about 1.5 at the next call, at fixed dwell. To move an agent off a stuck project, reset its context; do not expect a context crowded with other work to loosen it. Agents get stickier the longer they stay (leave odds fall by about 45% per e-fold of dwell), and the self-share does not explain it.

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RESULTS VS PREDICTIONS

prediction	observed	verdict
P1 $\beta_F > 0$ in $\geq 2/3$ units	3/25	fails
P2 β_F CI $\ni 1$; RE mean $\ni 1$	6/25; RE +0.07 [-0.05, +0.19]	fails
P3 $\varepsilon(F)$ above proxy band	+0.001 vs band 0.055	unpowered
P4 dwell tail follows (Kill A)	γ outside band 25/25	fails
P5 aging left over ($\beta_d < 0$)	G51 -0.58 [-0.62, -0.55]	holds
N1 reset step $\approx$ urn's placebo (20 calls ear-lier)	+0.40 [+0.25, +0.54] vs +0.016 [-0.19, +0.14]	fails
N2 same slope by goal type	difference +0.84 [+0.31, +1.37]	fails

GOAL PERIODS IN DETAIL

G36 × failed
G37 × failed
G38 × failed
G39 × failed
G40 ~ mixed
G41 × failed
G42 × failed
G44 × failed
G51 × failed
NE41 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Hops come from mentions: 27% are flickers ($A \rightarrow B \rightarrow A$ within 5 calls); without them the reset step is +0.46 (post hoc). Context segments last at most 41 calls, so on this skeleton even a true urn makes little dwell aging; $\varepsilon(F)$ and a null reset step were unpow-ered (synthetic power 0.45–0.65 and 0.52), and both were declared before the data. The card's forward tail test failed its size check (0.14–0.27) and was replaced by a posterior-predictive test (size 0.02–0.03). The richer self-share that counts the room's messages about the project gives larger slopes in G51 (+0.5 to +1.4); it is reported, not tested.

ROUND 2

Model the reset as a step clock (own calls since the last forced reset) and measure its decay. Separate intrinsic aging from per-visit frailty with repeat visits to one project. Explain why the self-share slope is positive in own-role weeks and negative in shared-goal weeks. Re-run with a minimum-dwell hop.

- **Essence.** where an agent's project commitment is stored, and which operator action releases it; round 1 says the con-text acts as a reset step, not as a self-share dose.
- **H134-R1** Reset as a step clock.
- **H134-R2** Aging vs frailty.
- **H134-R3** Goal-type slope.
- **H134-R4** Sustained hops.

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